

## Appendix A:

### Formal definitions

#### First Fundamental Form:

$$I = \begin{bmatrix} E & F \\ F & G \end{bmatrix} = \begin{bmatrix} 1+h_u^2 & h_u h_v \\ h_u h_v & 1+h_v^2 \end{bmatrix} \text{ with}$$

$$E = \vec{r}_u \cdot \vec{r}_u \quad F = \vec{r}_u \cdot \vec{r}_v \quad G = \vec{r}_v \cdot \vec{r}_v \quad \text{with } \vec{r} = (u, v, h[u, v]) \quad , \quad \vec{r}_u = \frac{\partial \vec{r}}{\partial u} \quad \text{and} \quad \vec{r}_v = \frac{\partial \vec{r}}{\partial v}$$

#### Second Fundamental Form:

$$II = \begin{bmatrix} L & M \\ M & N \end{bmatrix} = \frac{\begin{bmatrix} h_{uu} & h_{uv} \\ h_{uv} & h_{vv} \end{bmatrix}}{g} \quad \text{with} \quad g = \|\nabla h \cdot \nabla h^T\| = \sqrt{(1+h_u^2+h_v^2)}$$

$$L = \vec{r}_{uu} \cdot \vec{n} \quad M = \vec{r}_{uv} \cdot \vec{n} \quad N = \vec{r}_{vv} \cdot \vec{n}$$

then the coefficients of the ODE above are:

$$a = EM - FL, \quad b = EN - GL, \quad c = FN - GM$$

and Gaussian and Mean Curvatures are:

$$K = \frac{LN - M^2}{EG - F^2} \quad \text{and} \quad H = \frac{GL - 2FM - EN}{2(EG - FL)}$$

Principal Curvatures  $\kappa_1 \kappa_2$  along the principal directions forming the lines of curvature for a function  $h(u, v)$  in Cartesian coordinates.

$$\kappa_1 = \frac{h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1+h_v^2+h_u^2)^{\frac{3}{2}}} - \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1+h_v^2+h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1+h_v^2+h_u^2)^2}}$$

$$\kappa_2 = \frac{h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1+h_v^2+h_u^2)^{\frac{3}{2}}} + \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1+h_v^2+h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1+h_v^2+h_u^2)^2}}$$

Note that  $\kappa_1 \kappa_2 > 0$  if the Hessian is positive definite:  $h_{uu} h_{vv} - h_{uv}^2 > 0$

Monge/Local Astigmatism  $\kappa_1 - \kappa_2$

$$A = 2 \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2 h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1 + h_v^2 + h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1 + h_v^2 + h_u^2)^2}}$$

Mean curvature  $(k_1 + k_2)/2$

$$H = \frac{h_{vv} + h_{vv} h_u^2 - 2 h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1 + h_v^2 + h_u^2)^{\frac{3}{2}}}$$

Gaussian Curvature  $K = \kappa_1 \kappa_2$

$$K = \frac{h_{vv} h_{uu} - h_{uv}^2}{(1 + h_v^2 + h_u^2)^2}$$

The principal curvatures are also expressible as

$$\kappa_1, \kappa_2 = H \pm \sqrt{H^2 - K} = H \pm A/2$$

In polar coordinates this gets more cumbersome, though the ubiquitous denominator of first derivatives (the metric) is quite simple, so for  $r > 0$ :

$$h_u = h_r \cos(t) - \sin(t) \frac{h_t}{r}$$

$$h_v = h_r \sin(t) + \cos(t) \frac{h_t}{r}$$

$$(1 + h_v^2 + h_u^2) = \left(1 + h_r^2 + \left(\frac{h_t}{r}\right)^2\right)$$

The conversion for the second derivatives (Hessian) is

$$h_{uu} = h_{rr} - \sin^2(t) \left( h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) + 2 \cos(t) \sin(t) \left( \frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

$$h_{vv} = h_{rr} - \cos^2(t) \left( h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) - 2 \cos(t) \sin(t) \left( \frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

$$h_{uv} = \cos(t) \sin(t) \left( h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) + (\sin^2(t) - \cos^2(t)) \left( \frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

Curiously, if at the origin with  $h_t \neq 0$  then as  $r \rightarrow 0$ ,  $K \rightarrow \left( \frac{h_{tt}}{h_t} \right)^2$

Of course the ratio of  $h_t$  of to  $h_{tt}$  cannot be a function of  $t$  at  $r=0$ , because that would imply that  $K$  is not single valued at the origin. One way to achieve independence from  $t$  is if the surface is axisymmetric, then at the origin:

$K = (h_{rr})^2$ ,  $H = h_{rr}$  as follows from the axisymmetric versions of the equations below.

Simplified formulae for surfaces of revolution (axisymmetric surfaces with elevation  $z(r)$  where  $r$  is the radius from the axis of revolution) and all derivatives in  $\theta$  are zero:

$$z_\theta = 0, \quad z_{r\theta} = 0, \quad \text{and} \quad z_{\theta\theta} = 0$$

"Instantaneous or Tangential" curvature, polar coordinates:

$$\kappa_1 = \frac{z_{rr}}{(1 + z_r^2)^{3/2}}$$

which actually the formula for the first principal curvature for a surface of revolution (axisymmetric) parameterized by  $z=z(r)$  The other principal curvature in that case is actually the

"Axial or Sagittal" curvature ", polar coordinates:

$$\kappa_2 = \frac{z_r}{r \sqrt{1 + z_r^2}}, \text{ with } z_r \rightarrow 0 \text{ at } r \rightarrow 0 \text{ implies } \kappa_2 \rightarrow z_{rr}$$

Note also that the tangential curvature is the derivative of the sagittal curvature in this case.

As an example, consider a simple paraboloid,  $z(r)=r^2$ . The ratio between  $\kappa_1/\kappa_2 = (1+4r^2)$ , showing that for  $r \ll 1$ , the two are similar, but as the radius increases, the axial curvature becomes larger than the tangential curvature. Since the two principal curvatures are orthogonal, that also allows us to realize that the axial curvature is in the radial direction, whereas the tangential curvature is in the circumferential direction.

Mean Power is the arithmetic average  $(\kappa_1 + \kappa_2)/2$ .  $H$  is traditionally used.

$$H = \frac{1}{2} \left[ \frac{z_r}{r \sqrt{1+z_r^2}} + \frac{z_{rr}}{(1+z_r^2)^{3/2}} \right]$$

Gaussian Power is the product  $k_1 k_2$  of the principal curvatures. It's unfortunate that it isn't the geometric mean instead of its square...

$$K = \frac{z_r z_{rr}}{r(1+z_r^2)}$$

So in order to get a comparable "power" in diopters, the square root of K is used.

So called Monge Astigmatism  $\kappa_1 - \kappa_2$  :

$$A = \left[ \frac{z_r}{r \sqrt{1+z_r^2}} - \frac{z_{rr}}{(1+z_r^2)^{3/2}} \right]$$

Warp/Oblique power, same as axial but with  $z_\theta/r$  instead of  $z_r$   
This last definition is non-standard and defined here as:

$$K = \frac{z_\theta}{r^2 \sqrt{1+(z_\theta/r)^2}},$$

The Cartesian equation for the lines of curvature is given as the solutions of the ODE:

$$\frac{v'(t)^2 \left( h_v h_{vv} h_u - h_{uv} - h_v^2 h_{uv} \right)}{\sqrt{1+h_v^2+h_u^2}} + \frac{u'(t) v'(t) \left( h_{vv} + h_{vv} h_u^2 - h_{uu} - h_v^2 h_{uu} \right)}{\sqrt{1+h_v^2+h_u^2}} + \frac{u'(t)^2 \left( h_{uv} + h_u^2 h_{uv} - h_v h_u h_{uu} \right)}{\sqrt{1+h_v^2+h_u^2}} = 0$$

with  $h_u = \frac{\partial h}{\partial u}$ ,  $h_v = \frac{\partial h}{\partial v}$  which has two solution sets as the solution to  $u'(t) = m(u(t), v(t)) v'(t)$  where  $u(t)$ ,  $v(t)$  describe parametric curves in  $t$  using the two roots of the quadratic solution with some rearrangement  $m(u, v)$ :

$$m(u, v) = (-b - \text{sign}(b) \sqrt{b^2 - 4ac}) / 2a$$

or

$$m(u, v) = 2c / (-b - \text{sign}(b) \sqrt{b^2 - 4ac})$$

with

$$a = \left( \frac{(1 + h_u^2) h_{uv}}{\sqrt{1 + h_v^2 + h_u^2}} - \frac{h_v h_u h_{uu}}{\sqrt{1 + h_v^2 + h_u^2}} \right)$$

$$c = \left( \frac{h_v h_u h_{vv}}{\sqrt{1 + h_v^2 + h_u^2}} - \frac{(1 + h_v^2) h_{uv}}{\sqrt{1 + h_v^2 + h_u^2}} \right)$$

$$b = \left( \frac{h_{vv} (1 + h_u^2)}{\sqrt{1 + h_v^2 + h_u^2}} - \frac{(1 + h_v^2) h_{uu}}{\sqrt{1 + h_v^2 + h_u^2}} \right)$$

The "lines of curvature" describing the directions of the principal curvatures can be plotted as a vector field in  $(u, v)$ . The lines of curvature in the axisymmetric case are given by the meridians and circles of longitude.

The "Axial Power" is not defined for off center or non-axisymmetric case, but there are formal alternatives, for example, defining the following, based on the surface vectors and normals:

$\vec{v}=(u,v,h(u,v))$  with normal  $\vec{n}=\frac{(-h_u,-h_v,1)}{\sqrt{(1+h_u^2+h_v^2)}}$  and at the origin

the values  $\vec{v}_0=\vec{v}(0,0)$  and  $\vec{n}_0=\vec{n}(0,0)$  then we can define the vectors

$\vec{ac}=\vec{n}_0\times(\vec{v}-\vec{v}_0)$  and  $\vec{an}=\vec{n}\times\vec{n}_0$  which lead to one of three "radii"

$$\text{Definition 1}=\sqrt{\frac{\vec{ac}\cdot\vec{ac}}{\vec{an}\cdot\vec{an}}}=\frac{|\vec{ac}|}{|\vec{an}|}$$

$$\text{Definition 2}=\frac{\vec{ac}\cdot\vec{an}}{\sqrt{\vec{an}\cdot\vec{an}}}=\frac{\vec{ac}\cdot\vec{an}}{|\vec{an}|}$$

$$\text{Definition 3}=\sqrt{\frac{\vec{ac}\cdot\vec{ac}}{(\vec{n}_0\times\vec{X})\cdot(\vec{n}_0\times\vec{X})}} \text{ with } \vec{X}=\frac{(\vec{ac}\times\vec{n})\times\vec{ac}}{[(\vec{ac}\times\vec{n})\times\vec{ac}]}$$

or

$$\text{Definition 3}=\frac{|[(\vec{ac}\times\vec{n})\times\vec{ac}]||\vec{ac}|}{|(\vec{n}_0\times((\vec{ac}\times\vec{n})\times\vec{ac}))|}$$

(Klein SA, Mandell RB, Barsky BA. Representing corneal shape. )

Picking Definition 1, with  $\vec{n}_0=(0,0,1)$  which corresponds to no axial angular misalignment, only (u,v) misalignment, leads to

$$\kappa = \left( \frac{\sqrt{(h_u^2 + h_v^2)}}{\sqrt{((v - v_0)^2 + (u - u_0)^2)(1 + h_v^2 + h_u^2)}} \right)$$

which is very similar to the polar coordinate axisymmetric version

given earlier:  $\kappa=\frac{h_r}{r\sqrt{1+h_r^2}}$

Note that with  $\vec{n}_0=(0,0,1)$ , that the tangent plane of the surface is orthogonal to the VK axis as well as centered. Definition 1 explicitly assumes that rays stay in the meridional plane.

Picking Definition 2, with  $\vec{n}_0=(0,0,1)$  which also corresponds to no axial angular misalignment, only (u,v) misalignment, but with a different geometric interpretation, leads to

$$\kappa = \left( \frac{\sqrt{(h_v^2 + h_u^2)}}{(h_v(v - v_0) + h_u(u - u_0))\sqrt{(1 + h_v^2 + h_u^2)}} \right)$$

while Definition 3 (still with  $\vec{n}_0=(0,0,1)$ ) is quite complicated.